

Data

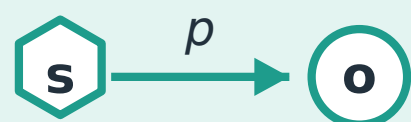
Documents



article passages

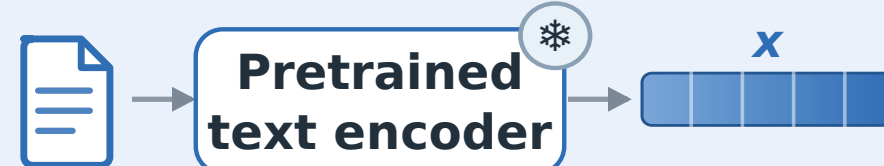
$\triangle$ same biomedical fact

KG triple



(subject, predicate, object)
chemical-gene fact

Data Representations



Triple composition φ

combine node embeddings into a triple embedding



concat
[s ; p ; o]

Hadamard
 $s \odot p \odot o$

L1 $|s + p - o|$

L2 $(s + p - o)^2$

Alignment

Projection head $g\theta$

Linear

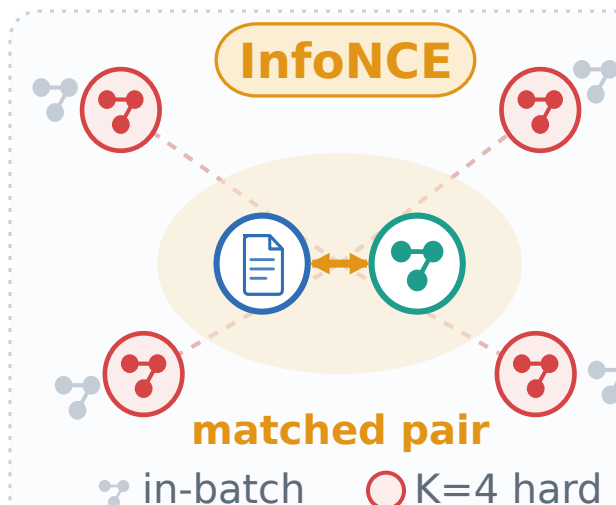
MLP

Cross-attn

Training direction



Shared embedding space



Hard negatives

via triple corruption

positive	(s, p, o)
$\times p$	(s, p', o)
$\times o$	(s, p, o')
$\times s$	(s', p, o')
$\times s, o$	(s', p, o')

Evaluation — bidirectional retrieval

D2T document $\rightarrow$ triple



retrieve the correct triple for a given document

T2D triple $\rightarrow$ document



retrieve the correct document for a given triple

❄ frozen encoder ⚡ trained projection 📄 document 🟩 KG triple 🟡 correct match